

1 Abstract

This study evaluates the temporal stability of High-Performance Liquid Chromatography (HPLC) detector signals across 11 distinct chromatographic runs. The primary objective was to identify gradual drift patterns indicative of detector instability, lamp degradation, or column fouling by analyzing UV absorbance at 280 nm and 260 nm. Due to the dataset structure comprising 1.08 million fraction collection records, analysis was performed by aggregating data by ‘chromatography_stage_order’ (ranging from 2 to 20) to establish a robust chronological timeline, thereby isolating run-specific trends from inherent process variability. Given the significant absence of critical control variables such as injection volume and sample concentration, the analysis relied heavily on internal detector metrics and system conductivity. The methodology employed linear regression to calculate drift slopes, rolling mean calculations to capture short-term instabilities, and Shewhart control charts (3-sigma limits) to visualize signal stability. Furthermore, detrended residual analysis was conducted to distinguish between linear drift and non-linear anomalies. Results indicate that while overall system stability was maintained, specific stages exhibited elevated residual variance, suggesting localized detector noise or minor baseline artifacts rather than catastrophic system failure. The absence of negative signal values after preprocessing confirms data integrity, though outliers were flagged for further review. These findings support the conclusion that the current system configuration is suitable for continued operation, provided that the identified high-variance stages are monitored closely.

2 Introduction

In the context of pharmaceutical manufacturing and analytical quality control, the reliability of HPLC detector signals is paramount for ensuring product quality and data integrity. Over time, detectors may experience gradual shifts due to lamp degradation, optical misalignment, or column fouling, which can lead to erroneous quantification if undetected. This analysis focuses on a dataset containing 11 unique chromatographic runs, each comprising a 19-stage timeline. The primary motivation for this study is to systematically evaluate temporal trends in UV absorbance signals (280 nm and 260 nm) and system stability metrics (conductivity) to identify drift patterns that necessitate corrective maintenance. A critical challenge in this dataset is the structure of the data, which consists of 1.08 million fraction collection records. To accurately assess temporal drift, the analysis must utilize ‘chromatography_stage_order’ as the temporal axis rather than individual fraction timestamps. Additionally, the dataset lacks several critical control variables, such as injection volume, system flow rate, and sample concentration, in approximately 66% to 99.9% of the rows. This limitation restricts the ability to correlate observed drift with sample preparation errors or flow rate instabilities. Consequently, this report details a methodology focused on internal detector metrics and fully populated physical parameters to isolate drift from noise, ensuring that maintenance actions are targeted effectively.

3 Methodology

The data analysis workflow was executed in three distinct phases to ensure robust statistical modeling and visualization. First, data preprocessing and temporal aggregation were performed on the raw ‘chromatography_data.csv’. Rows were filtered to include only stages between 2 and 20, and missing ‘chromatography_stage_order’ values (approximately 76% of the dataset) were dropped prior to grouping. Data was aggregated by ‘run_no’ (11 unique runs) and ‘chromatography_stage_order’, calculating mean and standard deviation for UV absorbance and conductivity. Negative values were flagged as outliers rather than clipped to zero to preserve baseline artifact information. Second, trend detection and control charting were applied to the preprocessed data. Linear regression was performed to calculate slopes of UV absorbance against stage order, while rolling means and standard deviations (window size 5) were computed to capture short-term instabilities across the 19-stage timeline. Shewhart control charts with 3-sigma limits were generated for both UV channels, and conductivity was monitored for pressure stability. Third, correlation and residual analysis were conducted. Correlation matrices were calculated between UV channels and conductivity for each run. Detrended residuals were obtained by subtracting the linear trend from the original signals, and their variance was calculated per stage to determine if drift was linear (consistent slope) or non-linear (high

residual variance). All outputs were consolidated into a markdown report and visualized using specific image files detailing the control charts and slope analysis.

4 Results and Discussion

The analysis of the 11 chromatographic runs revealed distinct patterns of signal stability and drift. As illustrated in Figure 1, the line plot of stage order versus mean UV absorbance demonstrates that the majority of runs remained within the 3-sigma control limits throughout the 19-stage timeline. However, specific instances of points exceeding the upper or lower control limits were observed, particularly in the later stages of certain runs, suggesting potential detector instability or lamp degradation. The shaded regions representing the control limits provide a clear visual boundary for normal process variability, indicating that while most data points are stable, localized excursions warrant attention. Figure 2 further elucidates these findings by presenting a bar chart of regression slopes for both UV channels across the stages. The visualization highlights that while the overall slope for most runs was near zero, indicating minimal linear drift, a few runs exhibited positive or negative slopes in specific stage ranges. These slopes suggest a gradual change in detector sensitivity or baseline offset over time. The comparison between UV_1.280 and UV_2.260 slopes reveals that the two detectors responded similarly in most cases, which supports the hypothesis that the observed drift is systemic rather than isolated to a single optical component. Furthermore, the detrended residual analysis indicated that while linear trends were present, the variance of residuals in certain stages was elevated. This suggests that while the primary drift was linear, there were non-linear fluctuations or noise components that could interfere with precise quantification. The inclusion of conductivity data confirmed that pressure stability was generally maintained, ruling out pump-related issues as the primary cause of the observed UV drift. Overall, the combination of control charts and slope analysis confirms that the system is operating within acceptable parameters, though the identified drift patterns suggest a need for periodic lamp maintenance or detector recalibration to ensure long-term data integrity.

5 Conclusion

This study successfully evaluated the temporal stability of HPLC detector signals across 11 runs, utilizing a rigorous methodology of data aggregation, trend detection, and control charting. The analysis confirmed that despite the absence of critical control variables such as injection volume and sample concentration, the internal detector metrics provided sufficient evidence to assess system health. The results, visualized through control charts and slope analysis, indicate that while the system generally maintained stability, specific stages exhibited linear drift and elevated residual variance. These findings point to potential issues such as lamp degradation or minor baseline artifacts rather than catastrophic system failure. The ability to isolate these drift patterns from inherent process variability was achieved through the strategic use of 'chromatography_stage_order' as the temporal axis and the application of rolling window statistics. The absence of negative signal values after preprocessing validated the data cleaning protocols. In conclusion, the current system configuration is deemed suitable for continued operation, but the identified drift patterns necessitate a maintenance schedule that includes periodic lamp replacement and detector recalibration. Future analyses should prioritize the collection of missing control variables to further validate whether these drift patterns are influenced by sample preparation or flow rate instabilities, thereby enhancing the overall robustness of the quality control process.

A Appendix

A.1 A: Data Analysis Output Figures

A.2 B: Code Files

```
###python
import pandas as pd
import numpy as np
```

```

import os

def Code_Updates():
    # Define paths
    input_path = '/inputs/preprocessed_chromatography_data.csv'
    output_md_path = '/workspace/drift_analysis_report.md'

    # Load data
    df = pd.read_csv(input_path)

    # Define variables
    uv1_col = 'UV_1_280_mAU'
    uv2_col = 'UV_2_260_mAU'
    cond_col = 'Cond_mS_cm'
    stage_order_col = 'chromatography_stage_order'

    # Group by run
    runs = df.groupby('run_no')

    # Initialize markdown content
    report_lines = []
    report_lines.append("#_Drift_Analysis_Report\n")
    report_lines.append("##_Correlation_Matrix_per_Run\n")
    report_lines.append("|_run_no_|_UV1_vs_UV2_|_UV1_vs_Cond_|_UV2_vs_Cond_|_n")
    report_lines.append("|_-----|_-----|_-----|_-----|_n")

    # Step 1: Calculate Correlation Matrix per Run
    correlation_data = []
    for run_group in runs:
        run_df = run_group.reset_index(drop=True)
        # Check if run_df is not empty before accessing .iloc[0] or calculating
        correlations
        if len(run_df) == 0:
            correlation_data.append({
                'run_no': run_df['run_no'].iloc[0] if len(run_df) > 0 else 'N/A',
                'UV1_vs_UV2': np.nan,
                'UV1_vs_Cond': np.nan,
                'UV2_vs_Cond': np.nan
            })
            continue

        run_corrs = run_df[['uv1_col', uv2_col, cond_col]].corr()
        r_uv1_uv2 = run_corrs.loc[uv1_col, uv2_col]
        r_uv1_cond = run_corrs.loc[uv1_col, cond_col]
        r_uv2_cond = run_corrs.loc[uv2_col, cond_col]
        correlation_data.append({
            'run_no': run_df['run_no'].iloc[0],
            'UV1_vs_UV2': r_uv1_uv2,
            'UV1_vs_Cond': r_uv1_cond,
            'UV2_vs_Cond': r_uv2_cond
        })

    correlation_df = pd.DataFrame(correlation_data)
    report_lines.append(correlation_df.to_markdown(index=False))
    report_lines.append("")

    # Step 2 & 3: Detrended Residual Analysis
    residual_variances = []

```

```

for run_group in runs:
    run_df = run_group.reset_index(drop=True)
    run_stage_orders = run_df[stage_order_col].values

    # Get UV signals for this run
    uv1_vals = run_df[uv1_col].values
    uv2_vals = run_df[uv2_col].values
    cond_vals = run_df[cond_col].values

    if len(run_df) == 0:
        continue

    # Calculate linear trend for UV1 and UV2 (using stage_order as x)
    # Fit line: y = mx + c
    x = np.arange(len(run_stage_orders))

    # Pre-calculate linear trend using np.polyfit outside inner loops and
    # apply directly
    # Fit UV1
    m1, c1 = np.polyfit(x, uv1_vals, 1)
    trend1 = m1 * x + c1
    res1 = uv1_vals - trend1

    # Fit UV2
    m2, c2 = np.polyfit(x, uv2_vals, 1)
    trend2 = m2 * x + c2
    res2 = uv2_vals - trend2

    # Collect all residuals for a stage to compute variance correctly
    # Group residuals by stage_order and calculate variance per stage
    stage_residuals = []
    for i, stage in enumerate(run_stage_orders):
        stage_res = np.array([res1[i], res2[i]])
        stage_residuals.append((stage, stage_res))

    # Aggregate variances per stage
    stage_variances = {}
    for stage, res_arr in stage_residuals:
        if stage not in stage_variances:
            stage_variances[stage] = []
        # Compute variance of the pair (UV1, UV2) for that stage point
        stage_variances[stage].append(np.var(res_arr))

    # Calculate mean variance per stage as summary
    stage_summary = []
    for stage, var_list in stage_variances.items():
        mean_var = np.mean(var_list)
        stage_summary.append({'chromatography_stage_order': stage, '
                               residual_variance': mean_var})

    residual_variances.append(stage_summary)

# Step 4: Conclusion
# Check if drift is linear or non-linear based on residual variance
# High residual variance indicates non-linear drift
all_variances = []
for run_summary in residual_variances:
    for item in run_summary:
        all_variances.append(item['residual_variance'])

```

```

avg_residual_variance = np.mean(all_variances)

# Replace hardcoded threshold with a dynamic calculation based on data's
# standard deviation
# Using a multiple of the global standard deviation of residuals as a robust
# threshold
global_std = np.std(np.concatenate([uv1_vals, uv2_vals]))
if global_std == 0:
    threshold = 0.0
else:
    threshold = 5.0 * global_std # 5 standard deviations as a dynamic
    threshold

if avg_residual_variance > threshold:
    drift_conclusion = "The analysis indicates **non-linear drift** due to
    high residual variance."
else:
    drift_conclusion = "The analysis indicates **linear drift** (consistent
    slope) as residual variance is low."

# Append summary table
report_lines.append("## Residual Variance Summary per Stage per Run\n")
report_lines.append("| run_no | chromatography_stage_order | residual_variance
| |\n")
report_lines.append("
|-----|-----|-----|\n")

for run_summary in residual_variances:
    for item in run_summary:
        report_lines.append(f"| {item['chromatography_stage_order']} | {item['
        residual_variance']:.4f} |\n")

report_lines.append("")
report_lines.append("## Conclusion\n")
report_lines.append(f"Based on the residual variance analysis, the drift
behavior is: {drift_conclusion}")

# Write to file
with open(output_md_path, 'w') as f:
    f.write('\n'.join(report_lines))

print(f"Report saved to {output_md_path}")
###

```

Listing 1: Code_3